

Cole Foos

Software Engineer · Distributed Systems & Authorization

Norwalk, Ohio | cole@foos.dev | [linkedin.com/in/colefoos](https://www.linkedin.com/in/colefoos)

SUMMARY

Software engineer with 5+ years building large-scale, high-throughput distributed systems, now focused on identity and authorization at AWS. Designs and ships production Rust services with single-digit-millisecond latency and deep platform integrations, leading cross-team design while staying hands-on across the full stack. Seeking a mid-to-senior backend / distributed-systems role.

SKILLS

Languages: Rust, Java, Python, TypeScript, C#, SQL, React, Angular

Cloud & Infra: AWS, Docker, Distributed Systems, CI/CD, Caching, S3 / Athena / Glue

Domains: Authorization & Identity, Zero-Trust, Cedar, Machine Learning, Large-Scale Services

EDUCATION

B.S. Computer Science

University of Toledo

Aug 2016 – Dec 2020 · Cum Laude

Regent's Award · UT Writing Excellence Award · Dean's List

EXPERIENCE

Software Development Engineer III — Amazon Web Services Apr 2025 – Present

Amazon Verified Permissions (AVP) — managed authorization service · Arlington, VA

- Delivered **FIPS, PrivateLink, and IPv6 support** for Amazon Verified Permissions, expanding the service's compliance reach and secure connectivity for enterprise and regulated customers.
- Built integrations between AVP and **AWS IAM Identity Center**, giving AWS first-party applications a seamless, unified path to externalized authorization.
- Lead the team in designing **internal authorization-query APIs** (slated for external launch) that let service teams answer questions like “who has access to what?” and “what resources can a user reach?”
- Authored a **Cedar language server**, the latest open-source contribution to the Cedar authorization ecosystem, improving the editing experience for policy authors.

Software Development Engineer II — Amazon Web Services Nov 2023 – Apr 2025

Zero-trust identity service · Arlington, VA

- Designed and built a **Rust-based managed authentication/authorization service** powering a new zero-trust identity offering.
- Engineered caching that achieved **single-digit-millisecond response times** at scale.
- Vendored internal Rust libraries adopted by **other AWS service teams**, accelerating their integration and reducing duplicate work.
- Drove cross-team design, breaking work into clear task plans that aligned multiple teams on implementation.

Software Development Engineer II — Amazon Oct 2022 – Nov 2023

Lifecycle marketing platform · Arlington, VA

- Engineered a **drag-and-drop lifecycle marketing tool** enabling marketers to personalize experiences for **millions of users per day**.
- Developed a cross-team **ML feature** that automatically selects the best-performing customer experience.
- Cut **S3, Athena, and Glue** resource utilization through storage reductions, query optimization, and partitioning.
- Built an automated system to halt under-performing or malfunctioning campaigns; mentored teammates and participated in on-call.

Software Development Engineer I — Amazon May 2021 – Oct 2022

Lifecycle marketing platform · Remote

- Designed a notification system delivering **hundreds of weekly alerts** on campaign progress, engagement, and issues.
- Built a high-capacity event processor handling **millions of events** to engage users at key lifecycle milestones.

EARLIER EXPERIENCE

Software Developer Intern — OnPoint Group Feb 2019 – Feb 2021

Perrysburg, OH / Remote

- Built an **invoice-automation system** (Amazon Textract, Rekognition, Salesforce) processing hundreds of invoices/day; tuned Azure Cosmos DB for a **10x reduction in page load times**.
- Delivered mobile-app backend services with a third-party vendor, a **CI/CD pipeline** in GitHub Actions, Power BI dashboards, and CRM integrations.

Software Development Engineer Intern — Amazon May 2020 – Aug 2020

Remote

- Designed and shipped a debugger tool letting marketers see a customer's position within their marketing campaigns.